What do you want to learn?

ALICE L WALTON SCHOOL OF MEDICINE

# Hemophilia

Last updated October 21, 2025 13 min read ScholarRx
**Adapted by:** Chelsey Deel
**Editor:** Jordan Finn
Adapted from: Hemophilia

Listen to this Brick
0:00 / 20:47 1x

Learning Objectives (4) Versions

*After completing this brick, you will be able to:*

1. Differentiate the clinical presentation of hemophilia A and B
2. Explain the pathophysiology and inheritance patterns of hemophilia A and B.
3. Interpret clinical findings to properly diagnose hemophilia A and B.
4. Outline the [illegible]

CASE CONNECTION

TR is a 6-year-old male brought to the emergency department by his parents because of a persistent nosebleed. "We've tried everything we know about stopping nosebleeds and nothing worked," TR's mom says. TR is adopted, and his family history is unknown. On exam, he has crusted and fresh blood in his nose and diffuse ecchymoses. After attending to TR's nosebleed, you tell his parents, "We need to do some bloodwork to see why he is having this bleeding problem." TR has a normal platelet count and normal PT/INR (PT- prothrombin time; INR- international normalized ratio); his PTT (partial thromboplastin time) is elevated and corrects when normal plasma is added to his sample.

What other testing will confirm your diagnosis? How will you explain this condition to TR's parents? Consider your answers as you read, and we'll revisit TR at the end of the brick.

GO TO CONCLUSION ↓

## What Is Hemophilia?

Hemophilia is a bleeding disorder resulting from defects in factors of the coagulation cascade, leading to decreased blood clotting, sometimes with severe bleeding episodes. The two primary types of hemophilia are A and B. Hemophilia A is a result of mutations in the factor VIII gene, whereas hemophilia B is a result of mutations in the factor IX gene. These mutations lead to decreased acti[illegible]rs, which then leads to an inability to acti[illegible] in fibrin formation. Because of the inability to form fibrin, the clot does not stabilize with the fibrin "glue," and excessive bleeding occurs, especially after trauma.

Most hemophilia cases are inherited, but about 30% of hemophilia A cases arise from spontaneous mutations in the factor VIII gene.

CLINICAL CORRELATION

Hemophilia C is an inherited disorder of factor XI but is extremely rare (~1% of hemophilia cases in the U.S.).

Defects in which coagulation factors cause hemophilia A and B?

### Epidemiology

Both types of hemophilia are rare, with only a combined frequency of about 1 in 5,000 live male births. Of the two types, hemophilia A is about four times more common.

## How Do Patients with Hemophilia Present (LO #1)?

The clinical features of hemophilia include easy bruising (**ecchymoses**), as well as uncontrolled hemorrhage after trauma or surgery.

Parts of the body that are subject to repeated mechanical stresses, such as the joints, often experience "spontaneous" hemorrhages (hemarthrosis) that can ultimately result in joint deformities (Figure 1).

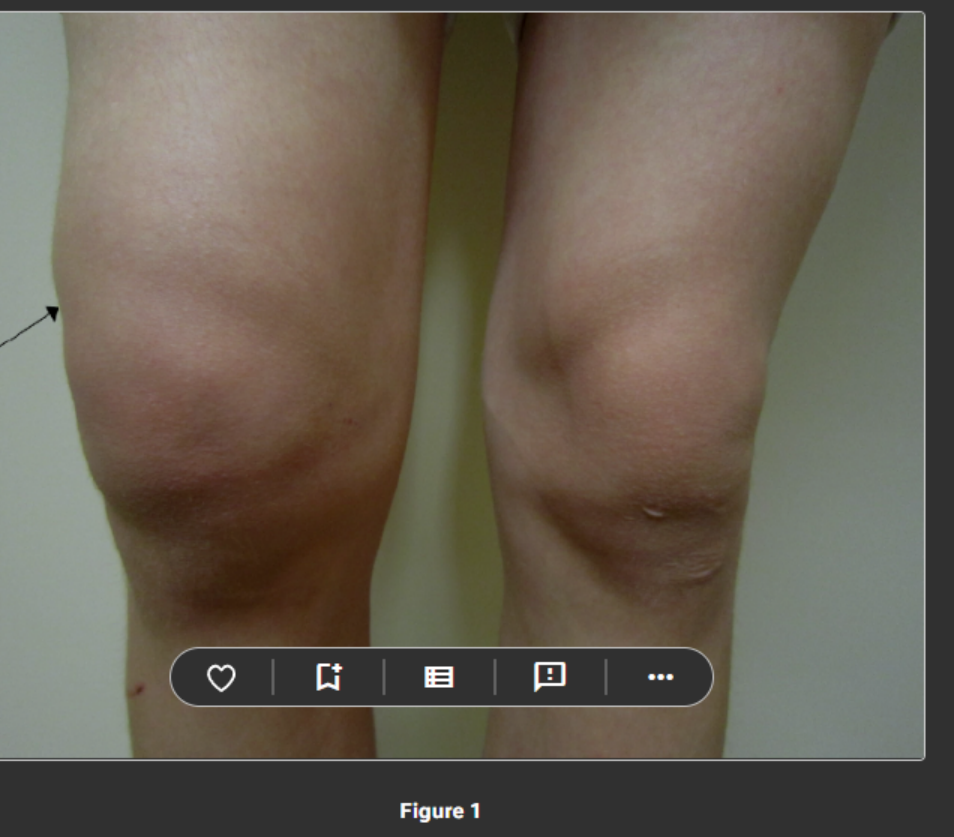

Figure 1

The most commonly affected joints are the knees, ankles, and elbows. Due to recurrent joint bleeding, the surrounding muscles, nerves, and underlying bones all suffer structural damage that leads to chronic pain and impaired mobility. The progressive worsening of the symptoms may eventually lead to joint replacement.

Bleeding can also occur from the genitourinary tract, leading to persistent hematuria, as well as the gastrointestinal tract, leading to blood in the stool. Additional sources of hemorrhage in hemophiliac patients include deep tissue bleeds (deep muscle, retroperitoneal) and intracranial bleeds (most common cause of death from hemorrhage for patients with hemophilia).

## What Is the Pathophysiology of Hemophilia (LO #2)?

We'll next discuss how the clotting factors are affected by hemophilia, then the genetic basis of the defects.

### Clotting Disorder

In the intrinsic arm of the coagulation cascade, activated factor VIII (VIIIa) serves as a cofactor for activated factor IX (IXa). They cooperate to activate factor X to Xa at the start of the common pathway, eventually leading to fibrin (Ia) formation, which [illegible]

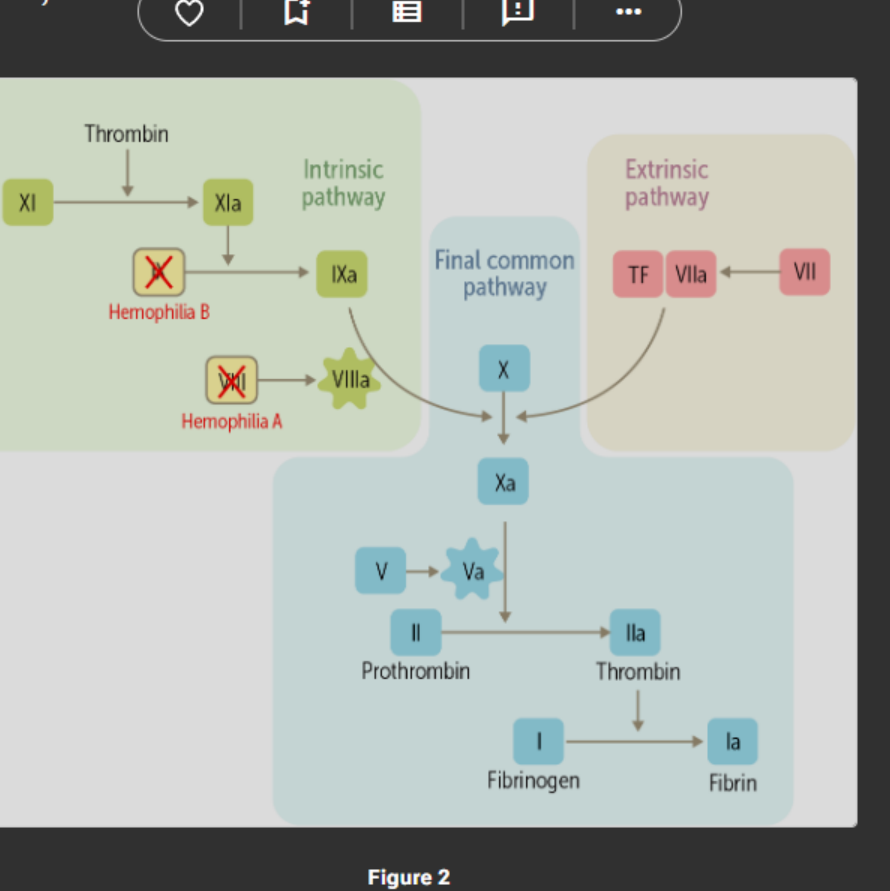

Figure 2

In patients who have decreased factor levels, those with symptoms will typically have less than 30% of the normal circulating levels of factors VIII or IX, and patients with severe disease will often have less than 1% of normal activity.

### Genetics

Many different mutati[illegible]use decreased levels of clotting factors, wh[illegible]ction (with normal levels). Disease severity varies widely.

Both hemophilia A and B are **X-linked recessive disorders** (Figure 3).

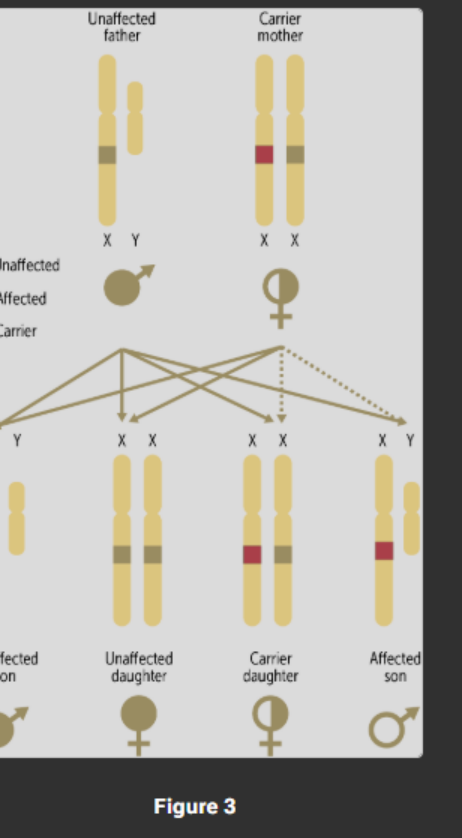

Figure 3

These disorders are most often seen in males because males have only one X chromosome. The disease is uncommon in females because they have a normal second X chromosome. However, hemophilia A can sometimes be seen in heterozygous females in the rare event that the X chromosome carrying the normal factor VIII ge[illegible] of cells (termed unequal lyonization).

Why are males more likely to have hemophilia?

CLINICAL CORRELATION

Hemophilia C is inherited in an autosomal recessive pattern.

## How Do We Diagnose Hemophilia (LO #3)?

The diagnosis, once suspected from history and physical, is made by coagulation tests and specific factor assays.

### Coagulation Testing

The activated partial thromboplastin time (aPTT) measures the intrinsic coagulation pathway, while prothrombin time (PT) tests the extrinsic coagulation pathway. Because hemophilia causes a defect in the intrinsic arm of the coagulation cascade, we expect to find a **prolonged aPTT**, reflecting the longer time for blood to clot in these patients. When normal plasma is mixed with the patient's blood sample, the missing clotting factors are restored and cause the PTT to return to normal in this test **(corrects with mixing study)**.

On the other hand, **PT is normal** in patients with hemophilia because the extrinsic arm of the coagulation cascade is unaffected.

Hemophilia B is clinically indistinguishable from hemophilia A on this routine coagulation testing. This makes sense because factor IX cooperates with factor VIII in the activation of factor X, and deficiencies of both factor IX and factor VIII result in impairment of the intrinsic arm of the coagulation cascade. So both forms show a prolonged PTT and a normal PT.

### Factor Assays

To differentiate hemophilia A and B, the diagnosis is made using **specific assays for factors VIII and IX**.

Table 1. Clinical Classification of Hemophilia A and B

| Factor Level | Bleeding Risk |
|---|---|
| Mild (6-30% activity) | Rare bleeding episodes- usually due to major trauma or surgery; may not be discovered until adulthood |
| Moderate (1-5% activity) | [illegible] problems (usually trauma-induced); can reach adulthood without crippling effects |
| Severe (<1% activity) | Many bleeding episodes, usually spontaneous; rarely reach adulthood without treatment |

### Differential Diagnosis

Disorders that impair platelet plug formation, such as von Willebrand disease, can be confused with hemophilia, but hemophilia patients lack petechiae, which result from bleeding caused by minor damage in capillary blood vessels. Hemarthroses (bleeding in the joints) are more common in hemophilia.

CLINICAL CORRELATION

von Willebrand disease is an autosomal dominant disorder caused by a deficiency of or defect in von Willebrand factor (vWF), which anchors platelets to damaged blood vessels and protects circulating factor VIII from degradation. Severe forms of the disease can strongly resemble hemophilia A because the lack of vWF decreases the stability of circulating factor VIII.

Why is PTT prolonged but PT unaffected in hemophilia?

## How Do We Treat Hemophilia (LO #4)?

Patients with bleeding disorders such as hemophilia should be monitored carefully for bleeding when they have surgical or major dental procedures, such as dental extractions. This might include extra testing, consulting a hematologist, and ensuring that clotting factors are available should problems arise.

Both hemophilia A and B are treated by replacing the deficient clotting factor with **recombinant factor infusions**. About 15% of patients with severe hemophilia A develop antibodies to recombinant factor VIII because their bodies recognize it as a foreign protein (this is called an **inhibitor**). In these patients, the prolonged PTT is **not** corrected when mixed with normal plasma because antibodies bind the free factor VIII, preventing its function.

CLINICAL CORRELATION

Inhibitor levels are measured in **Bethesda units**. The higher the number, the higher the titer of the [illegible] or low titer inhibitors, patients a[illegible] overwhelm the inhibitor. For moderate or high titer inhibitors, patients can be given recombinant factor VIIa (for acute bleeding as it directly forms Factor Xa and thrombin), activated prothrombin complex concentrates (bypass the inhibitor), or emicizumab (a monoclonal antibody against Factor IXa and X).

Desmopressin may be used to treat some patients with mild hemophilia A because it can stimulate the release of von Willebrand factor (vWF) from platelets, increasing factor VIII stability and leading to increased factor VIII levels. This treatment is effective in patients with hemophilia A with reduced factor VIII production, but it doesn't work in those who have reduced factor VIII function (with normal levels).

Why is desmopressin useful for the treatment of mild hemophilia A?

CLINICAL CORRELATION

Cryoprecipitate and fresh frozen plasma (FFP) are considered contraindicated for treatment of hemophilia A. Cryoprecipitate contains factor VIII but is not extensively treated to prevent infection (there is an infection risk). FFP does contain factor VIII but in a very low concentration, so you would have to give large amounts to have the same effect; additionally, FFP is also not extensively treated to prevent infection.

## Side Note: What is Acquired Hemophilia?

Acquired hemophilia arises when a *non-hemophiliac* patient acquires an antibody against specific coagulation factors. This is fundamentally a type of autoimmune disease. It is a rare disorder that tends to occur predominantly in older individuals, with a female predominance. It can occur in patients with malignancy, autoimmune diseases such as lupus, due to medications, or rarely post-partum; approximately half of cases are idiopathic- no known precipitating cause.

Antibodies to Factor VIII are most common, and patients present with secondary hemostasis-type bleeding, such as large hematomas, muscle bleeds, etc. (similar to Hemophilia A).

Treatment includes recombinant Factor VIIa or prothrombin complex concentrates for acute bleeding and immunosuppression to eliminate the inhibitor (prednisone +/- cyclophosphamide, rituximab).

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to TR, what other testing would confirm your diagnosis? How do you explain this condition to his parents?

An elevated partial thromboplastin time (PTT) that corrects with addition of normal plasma and a normal prothrombin time (PT) in a child with abnormal bleeding points to hemophilia A or B. You suspect A, a mutation in the factor VIII gene, because it is more common. A specific factor VIII assay comes back abnormal, confirming the diagnosis. You explain to TR's parents that he has a bleeding disorder due to a decrease in a specific factor that is essential for normal blood clotting. You refer TR to a hematologist to discuss treatment. TR does well over the next year with no episodes of serious bleeding.

## Summary

- Hemophilia is a bleeding disorder resulting from defects in the coagulation cascade, primarily due to mutations in factor VIII gene (Hemophilia A) or factor IX gene (Hemophilia B). Mutations in these factors prevent fibrin formation. Because of the inability to form fibrin, the clot does not stabilize with the fibrin "glue," and excessive bleeding occurs, especially after trauma.
- Most cases of hemophilia are inherited, but about 30% of Hemophilia A cases arise from spontaneous mutations.
- Hemophilia is more common in males due to its X-linked recessive inheritance pattern.
- Clinical features [illegible] uncontrolled hemorrhage after [illegible] hemorrhages often resulting in joint deformities.
- Diagnosis is made by coagulation tests and specific factor assays, with patients typically showing a prolonged activated partial thromboplastin time (aPTT) and normal prothrombin time (PT).
- Treatment involves replacing the deficient clotting factor with recombinant protein infusions, and in some cases of mild Hemophilia A, desmopressin may be used to stimulate the release of von Willebrand factor (vWF).
- Acquired hemophilia is an antibody that develops against a coagulation factor in a non-hemophiliac patient; it occurs predominantly in older female patients. Patients may have an underlying history of malignancy, autoimmune disease, or medication.

## Review Questions

**1. Deficiencies in which of the following best explains why both hemophilia A and B increase the activated partial thromboplastin time (aPTT)?**

A. ○ Activation of factor VII
B. ○ Fibrinolysis
C. ● Intrinsic arm of the coagulation cascade
D. ○ Platelet adhesion
E. ○ Stabilizing [illegible]

Hide Explanation

The correct answer is intrinsic arm of the coagulation cascade (C). Hemophilia A is a result of a defect in factor VIII, whereas hemophilia B is a result of a defect in factor IX. Both are important in the intrinsic arm of the coagulation cascade, which is tested by the partial thromboplastin time (PTT). Activation of factor VII (A) is incorrect because factor VII is part of the extrinsic coagulation cascade and is unaffected in hemophilia. Fibrinolysis (B) is the process of destroying the blood clot and is triggered by the activation of blood plasminogen; it occurs later, after the clot has outlived its usefulness. Platelet adhesion (D) and stabilizing factor VIII (E) are functions of von Willebrand factor.

**2. A patient with hemophilia B is planning to have children but is worried he will pass on his bleeding disorder. If his wife has no family history of bleeding disorders, and the couple has a son, what are the chances the son has the disease?**

A. ● 0%
B. ○ 25%
C. ○ 50%
D. ○ 100%
E. ○ Cannot be determined

Hide Explanation

The correct answer is 0% (A). Both hemophilia A and B are X-linked recessive disorders. The son's X chromosome would come from his unaffected mother, so he would have no chance of inheriting the disease from his father. Therefore, 25% (B), 50% (C), 100% (D), and cannot be determined (E) are incorrect. If the question asked about daughters, they would all receive his affected X chromosome and their mother's unaffected X chromosome, so 100% would be carriers.

**3. A patient comes to you complaining of easy bruising and bleeding that takes a long time to stop. Which finding suggests that the patient has a bleeding disorder other than hemophilia?**

A. ○ Family history of bleeding disorders
B. ○ Hemarthrosis
C. ○ Normal PT
D. ● Petechiae
E. ○ Prolonged PTT

Hide Explanation

The correct answer is [illegible]nd in disorders that impair platelet plug formation, such as von Willebrand disease. A family history of bleeding disorders (A), hemarthrosis (B), normal PT (C), and prolonged PTT (E) are common findings in hemophilia.

**4. What is an appropriate treatment for a patient with mild hemophilia A that experiences a minor bleeding episode?**

A. ○ Recombinant Factor IX
B. ● Desmopressin
C. ○ Recombinant Factor VIII
D. ○ Recombinant Factor VII
E. ○ Prothrombin complex concentrates

Hide Explanation

The correct answer is desmopressin (A). Desmopressin may be used to treat some patients with mild hemophilia A because it can stimulate the release of von Willebrand factor (vWF) from platelets, increasing factor VIII stability and leading to increased factor VIII levels. Recombinant Factor IX (A) is used to treat Hemophilia B. Recombinant Factor VIII (C) is used to treat more moderate-to-severe cases of Hemophilia A. Recombinant Factor VII (D) and prothrombin complex concentrates (E) can be used to treat Hemophilia A patients with a moderate to high titer inhibitor.

How would you rate this Brick?

## References

- Greer JP, et al, eds. *Wintrobe's Clinical Hematology*. 14th ed. Wolters Kluwer; 2018.
- https://www.nature.com/articles/s41572-021-00278-x
- https://www.uptodate.com/contents/clinical-manifestations-and-diagnosis-of-hemophilia?search=hemophilia&source=search_result&selectedTitle=1~150&usage_type=default&display_rank=1
- https://www.uptodate.com/contents/treatment-of-bleeding-and-perioperative-management-in-hemophilia-a-and-b?search=hemophilia&source=search_result&selectedTitle=2~150&usage_type=default&display_rank=2
- Kitchens CS, Kessler CM, Konkle BA, et al, eds. *Consultative Hemostasis and Thrombosis*. 4th ed. Elsevier; 2019.
- Kumar V, Abbas AK, Aster JC. Hematopoietic and Lymphoid Systems. In: Kumar V, Abas AK, Aster JC, eds. *Robbins and Cotran Pathologic Basis of Disease*. 10th ed. Elsevier; 2021.
- Salama M, ed. *Atlas* [illegible]vier; 2021.
search=hemophilia&source=search_result&selectedTitle=2~150&usage_type=default&display_rank=2
- Kitchens CS, Kessler CM, Konkle BA, et al, eds. *Consultative Hemostasis and Thrombosis*. 4th ed. Elsevier; 2019.
- Kumar V, Abbas AK, Aster JC. Hematopoietic and Lymphoid Systems. In: Kumar V, Abas AK, Aster JC, eds. *Robbins and Cotran Pathologic Basis of Disease*. 10th ed. Elsevier; 2021.
- Salama M, ed. *Atlas of Diagnostic Hematology*. 1st ed. Elsevier; 2021.

## Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Differentiate the clinical presentation of hemophilia A and B~~
02 ☑ ~~Explain the pathophysiology and inheritance patterns of hemophilia A and B.~~
03 ☑ ~~Interpret clinical findings to properly diagnose hemophilia A and B.~~
04 ☑ ~~Outline the management plan of hemophilia.~~

**Objective Confidence: 4/4**